

Can GPTs Evaluate Graphic Design Based on Design Principles?

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"Please rate between 1 to 10 points"

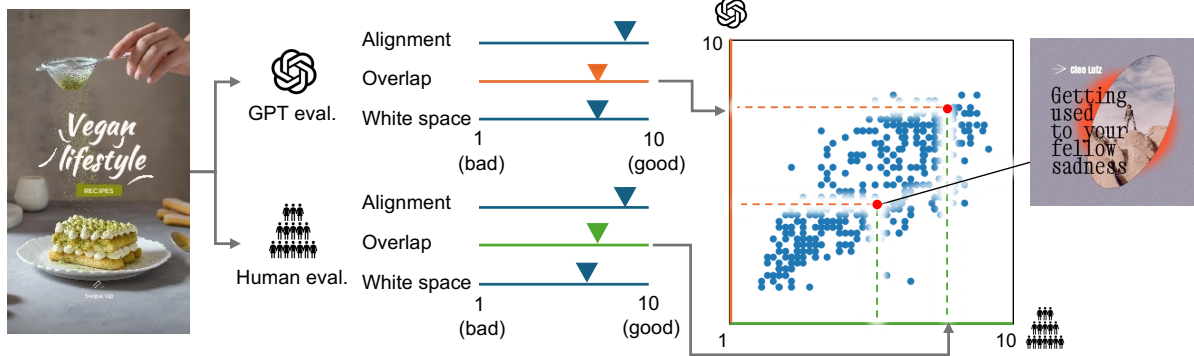


Figure 1: Overview of our study. We investigate GPT-based evaluation ability for graphic designs in three design principles “alignment,” “overlap,” and “white space.”

Abstract

Recent advancements in foundation models show promising capability in graphic design generation. Several studies have started employing Large Multimodal Models (LMMs) to evaluate graphic designs, assuming that LMMs can properly assess their quality, but it is unclear if the evaluation is reliable. One way to evaluate the quality of graphic design is to assess whether the design adheres to fundamental graphic design principles, which are the designer’s common practice. In this paper, we compare the behavior of GPT-based evaluation and heuristic evaluation based on design principles using human annotations collected from 60 subjects. Our experiments reveal that, while GPTs cannot distinguish small details, they have a reasonably good correlation with human annotation and exhibit a similar tendency to heuristic metrics based on design principles, suggesting that they are indeed capable of assessing the quality of graphic design. Our dataset is available at: <https://cyberagentailab.github.io/Graphic-design-evaluation/>.

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CCS Concepts

• Computing methodologies → Computer vision; Computer graphics.

Keywords

Large Multimodal Model, Large Language Model, Graphic Design, Graphic Design Evaluation

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1 Introduction

Foundation models learn from a large-scale corpus and exhibit remarkable generalization capability across various tasks, and the same is true for graphic design tasks [Chen et al. 2024; Cheng et al. 2024; Inoue et al. 2024; Jia et al. 2023]. While successful in certain graphic design tasks, it is still not apparent whether foundation models, such as GPT-4o, can reliably judge the quality of graphic design. Generally, high-quality graphic designs tend to follow design principles that are the designer’s common practices, such as alignment or repetition, as described in [Graham 2002; Williams 2014]. While humans can judge whether a given design follows these

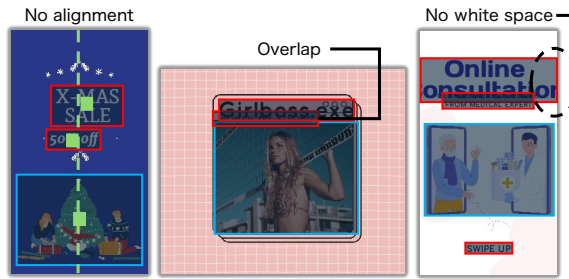


Figure 2: Negative examples of three design principles.

principles, human evaluation is time-consuming and not scalable. One of the early attempts in automatic evaluation is [O'Donovan et al. 2014], which introduces hand-crafted metrics for optimization. Recent studies of graphic design generation have employed Large Multimodal Models (LMMs), particularly GPT-4 [Achiam et al. 2023], to directly estimate the quality. However, these studies do not consider the viewpoints of design principles for the evaluation.

In this paper, we quantitatively study the behavior of GPTs with respect to human evaluators in assessing graphic designs, as illustrated in Figure 1. To investigate the performance of GPTs, we employ heuristic metrics as a baseline. We compare these approaches across three representative design principles, alignment, overlap, and white space, by manipulating graphic designs. We curate a dataset of graphic banner designs from an online service and perturb original designs to artificially generate low-quality designs for evaluation. Then, we ask human subjects to rate those designs in terms of the design principles. This human rate annotation allows us to assess which method – heuristic evaluation or GPTs – better aligns with human evaluation. We also discuss the qualitative comparison of heuristic and GPT-based evaluation.

We summarize our contributions in the following.

- We empirically study the relationship between hand-crafted and GPT-based evaluation for graphic designs with respect to human evaluation.
- We build a human-rated dataset of graphic designs that have varying degrees of quality to study the design quality.
- We find a higher correlation between human annotation and GPTs than heuristic metrics, suggesting that GPTs can give reliable judgment for graphic design quality under certain conditions.

2 Graphic design principles

Graphic design principles are the designer's common practices to create aesthetic work [Graham 2002; Williams 2014]. In this paper, we employ representative design principles used in several studies of graphic design generation [Kong et al. 2022; O'Donovan et al. 2014]: *alignment*, *overlap*, and *white space*. Alignment and overlap are commonly used in evaluation for layout generation, such as [Li et al. 2020]. White space is also a critical factor in graphic design generation approaches [Kong et al. 2022]. We follow three design principles based on [O'Donovan et al. 2014]. Below, we briefly describe each principle.

Alignment. We consider the arrangement of elements such that their edges line up along common rows or columns to express a sense of order and structure.

- (1) Alignment along with the horizontal and vertical direction is considered.
- (2) The elements that align at a glance but slight misalignment are penalized because it is visually displeasing.
- (3) Larger alignment groups (i.e., aligned elements distant from each other) are preferred as they produce simpler designs with more unity between elements.

Overlap. Inappropriate overlap reduces readability. We consider the following aspects.

- (1) The three types of overlap, the overlap of elements on text, the overlap of text on graphics, and the overlap of graphics on other graphics, are considered.
- (2) Hard-to-read text because of insufficient color contrast between a text and the background color is penalized.
- (3) The graphic design that includes elements extending past the boundaries is also penalized.

White space. White space is for the appropriate amount of space in a design for better readability.

- (1) A large ratio of white space that is not covered by design elements (e.g., graphics and texts) is preferred.
- (2) However, the graphic design with a too large region of empty white space on the image is undesirable.
- (3) The greater distance between each element is preferred.
- (4) Uniformed vertical spacing of each text element is preferred.
- (5) Wider border margins (i.e., the white space at the edges of the image) for each element are preferred.

For better understanding, we show some visually unappealing examples because they violate the principles in Figure 2.

3 Evaluation approach

In this paper, we compare the evaluation metrics of the heuristic approach and the GPT-based approach. In both approaches, we input a graphic design and then obtain the score of the input. We set the lower and upper bounds for both metrics, where the higher values indicate better quality.

3.1 Heuristic evaluation metrics

We employ the hand-crafted formulation in [O'Donovan et al. 2014] as heuristic evaluation metrics, which were originally designed for graphic design optimization. We adopt the formulas related to the alignment, overlap, and white space as evaluation metrics. These metrics directly consider the size and coordinates of text or other graphic elements. We normalize the metric range from 0 to 1.

3.2 GPT-based evaluation metrics

We give prompts to GPT-4o¹ and ask for scores in terms of alignment, overlap, and white space. We made the prompt based on explanations of design principles and formulas described in [O'Donovan et al. 2014]. See supplementary material for more details. We render and rasterize the original design to create an input prompt. We

¹<https://openai.com/index/hello-gpt-4o/>

Example of instruction:
Please rate between 1 to 10 points. Assess the graphic design in terms of [the name of design principle] from the following perspectives. [The design principles. (See subsection for each design principle)]



Figure 3: How to rate graphic designs by GPT and humans. The detailed input prompt for the GPT is described in the supplementary material.

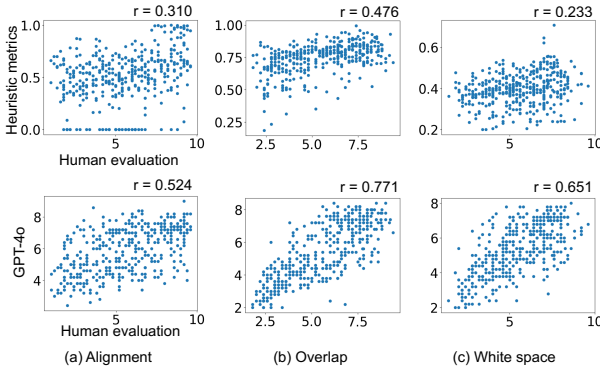


Figure 4: Correlation between human annotation and heuristic or GPT scores. The r is the Pearson correlation coefficient.

show an example in Figure 3, where GPT-4o gives scores from 1 to 10 to the input based on a specific aspect.

4 Dataset

We collect graphic design templates from VistaCreate², which hosts a large number of banner or poster designs. The templates contain coordinate and size information for graphic and text elements within a design. We randomly sampled one hundred templates from VistaCreate and perturbed coordinate and size parameters to create aesthetically inferior samples. We apply two kinds of perturbations in our experiments: x -coordinate and font size. We perturb the x -coordinate of the text position to evaluate alignment and the font size of text elements to evaluate overlap and white space. We give three ranges of perturbation, small, medium, and large, to investigate the sensitivity of metrics. We combined the 100 original samples and 600 perturbed samples to build a dataset of 700 samples.

For each design, we collect human scores for alignment, overlap, and white space. We recruited 60 participants via crowdsourcing and collected five annotations per graphic design, where we asked participants for scores in the 1 to 10 range, as shown in Figure 3. We use the average score of five annotations in our experiments.



Figure 5: Graphic designs with their alignment scores.

5 Experiments

5.1 Quantitative evaluation

Setup. For a fair comparison with human evaluation and detailed analyses, we conduct the GPT-based evaluation five times and use the average score for the GPT score. We set a sampling temperature, which controls the randomness of the GPT, to 1 (default value). Higher temperature (e.g., 0.9) shows more diverse output. Therefore, our experimental setting takes the diversity into account.

Correlation to human evaluation. To analyze the correlation between human annotation and the scores by automatic evaluation, we prepare two types of scatter plots of evaluation scores: one comparing heuristic metrics with human evaluation for each design principle, as shown in Figure 4. Interestingly, GPT-4o shows a better correlation with human evaluation than heuristic metrics. This may be because heuristic metrics are designed to optimize a graphic design by comparing the graphic design before and after optimization but not for comparing two totally different designs. We show an example of completely different graphic designs and their scores in Figure 5. According to the heuristic metrics, the order of better design is (a), (b), and (c). However, the order according to GPT-4o and human evaluation is (a), (c) (or (a) equal (c)), and (b). The heuristic evaluation quite degrades the score between (a) and (c) despite the fact that both graphic designs are created by human designers. On the other hand, GPT-4o and human evaluation are not much different. The results suggest that GPT-4o could be a suitable approach for quality evaluation for recent LMM-based generators since they generate significantly different designs by altering sampling parameters (or random seeds).

Sensitivity. We investigate whether GPT-based evaluation is effective across designs of diverse qualities by calculating the correlation coefficients for each perturbation level, as shown in Figure 6. GPT scores exhibit a stronger correlation with human annotation compared to heuristic scores across all metrics and perturbation levels. Additionally, the correlation of GPT scores tends to increase as perturbation levels rise. This implies that when perturbations are significant, such as in the case of a graphic design with a noticeably

²<https://create.vista.com>

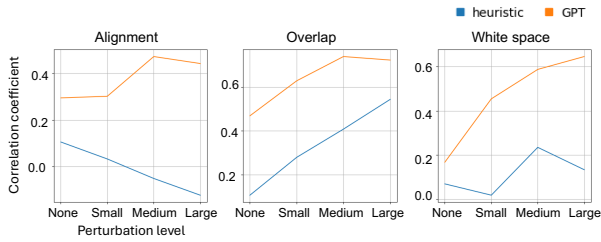


Figure 6: The Pearson correlation coefficient of the scores between human evaluation and each method.

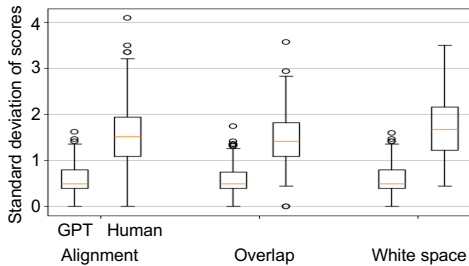


Figure 7: Box plot of the standard deviation of scores.

poor appearance, achieving a stable evaluation becomes easier and more efficient than humans.

Reliability. We investigate how stable the GPT scores are since running GPT multiple times may give different scores. We show the standard deviation of the GPT scores and human annotation in Figure 7. The standard deviation of GPT scores is lower than that of the human evaluation scores. This result suggests that GPT scores with a single run for each principle are practical and cost-effective evaluation metrics.

5.2 Qualitative analysis

We show typical cases where GPT-based evaluation succeeds while heuristic evaluation fails and vice versa. We show cases where only GPT-based evaluation succeeds in Figure 8. The design includes objects in the background, but heuristic evaluation considers the background as white space. Heuristic evaluation is difficult when assessing graphic designs that include an object embedded in the background.

We also show cases where only heuristic evaluation succeeds in Figure 9. Since heuristic metrics are vector-based, they can capture slight differences in the design directly from the vector values. However, GPT-based evaluation struggles to detect such differences accurately. A similar limitation of GPT has also been reported in another study [You et al. 2023].

6 Conclusion and Future Work

In this paper, we investigated the appropriateness of using heuristic evaluation metrics and GPT-4o for evaluating graphic designs focusing on design principles. To achieve this, we collected the large-scale human annotation and analyzed the correlation between the annotation and each evaluation metric. Our experiment showed that GPT-based evaluation better correlates with human annotation.

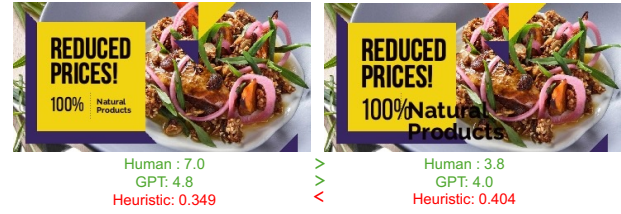


Figure 8: A sample with a correct white space assessment by GPT.

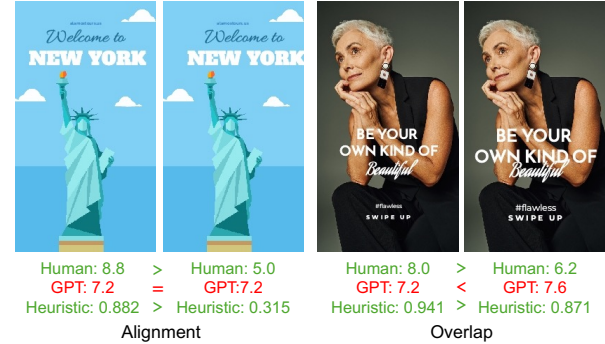


Figure 9: The examples of correctly assessed samples by heuristic evaluation.

For future work, we will jointly evaluate several design principles to assess the overall goodness of graphic designs. Additionally, we plan to evaluate graphic designs beyond design principles, such as font choices.

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